Physical Account: ShipOS

Basis: merged `fcd8c58` (PR #46). Annex A/B inputs and the assumption registry drive the public physics and D1-D4 engines. This is a simulated engineering prototype, not certification or a ship trial.

Eight resource budgets

Budget	Implemented model and material boundary
Power	Guest reserve at cruise: 200 kW, or 100 kW with one genset; two 150 kW mission lines. Peak reserve is +250/+100 kW for up to 60 s. Other speed reserves and lost-line selection are assumptions.
Heat	Water-cooled electrical load becomes loop heat 1:1. Capacity interpolates 160 kW at 12 °C to 120 kW at 30 °C, clamped outside that range; degraded factor × 0.6 gives 72 kW at 30 °C. Air-cooled G1 adds no loop heat. Inertia-buffer library is not a demonstrated transient decision model.
Deck	Declared discrete bay/hangar/roof slot groups and occupancy, not structural analysis.
Mast	Three positions, two occupied by ship equipment; declared guest position and mounting constraints.
Mass / stability	10 t raw margin; 12 t-equivalent proxy with mounting multipliers 1.0 bays, 1.2 hangar/roof, 3.0 mast. No GM, VCG, free-surface or hydrostatics.
Data links	Admission SATCOM worst case: 2 x 0.7 - 0.5 = 0.9 Mbit/s. Silence blocks SATCOM/radio; manoeuvring, deck machinery or layering blocks acoustic. Elastic losses are reported.
Compute	Guest allocation: 102 cores, 409.6 GB RAM, 3 whole accelerators (80% rack allocation).
Response time	A.8 ceilings and A.6 upper-bound transition planning; concurrent preparations with explicit dependencies. Acoustic hold precedes manoeuvring; human confirmation follows deck clearance. No measured latency compliance claimed.

Decisions, simulation and evidence

D1 tests the declared envelope corners; D2 ranks placements. D3/D4 are implemented. The opt-in controller executes supported chosen plans with crew gates and recorded operator input; observation mode applies no actions. API acceptance tests cover 14-crew recovery at t=841 after preparation, both priorities, reduced-crew denials and execution replay. Browser tests cover controls/rendering; synthetic edge tests are separate evidence. These do not establish physical safety or full challenge completion.

Accounting cautions and next constraint

Baseline B's t=720 trip is **not an overload**: observation nominal demand is 1.5 + 1 + 2 + 45 = 49.5 kW versus 100 kW. The 106.5 kW charging case requires an explicit overlap. G3 includes its cradle; surveying/holding/recovering draw 1 kW, charging 25 kW. G4 follows recovery -> launch -> flying, not charging. A loop rated 160 kW at 30 °C (**+40 kW**) is a proposed next investment, not an optimised or measured result. Cached forecasts and sea-state mapping are proxies.

Exact D4 metadata producer #43 remains separate; missing facts and ambiguous D3 deferrals fail closed. Winner-protection lifetime is provisional. G6 is absent. Deck-camera evidence is simulated. Replay needs compatible code and assumptions.

Registry appendix: 85 assumptions, 76 with literal uses, 9 without literal uses, plus one dynamic lookup, as generated at this baseline. The appendix records values, units, reasons, sources and use sites; unused structural choices are disclosed. The PDF contains this one-page account followed by that appendix.

Physical Account: assumption registry

Generated by `python -m tools.physical_account` from `data/assumptions/*.yaml`; do not edit. Narrative: PHYSICAL_ACCOUNT.md.

A *use* is a literal `A("<id>")` call in `shipos/` or `tools/` (Python) or a `"$<id>"` value in `data/` outside `data/assumptions/` (YAML). Tests and docs are not counted.

85 assumptions: 76 used, 9 unused; 1 dynamic lookup(s).

Dynamic lookups

Calls whose id is computed at run time; the ids they reach are not resolved here.

- `shipos/data.py:18` `A(node[len(REF):])`

Unused

No literal `A()` call or `$` reference reads these; some record modelling choices the code makes structurally.

- `data.satcom_admission_basis` (defined `data/assumptions/data_links.yaml:8`)
- `environment.sea_state_proxy` (defined `data/assumptions/environment.yaml:27`)
- `heat.capacity_interpolation` (defined `data/assumptions/heat.yaml:8`)
- `mass.margin_kept_t` (defined `data/assumptions/mass.yaml:3`)
- `positions.bay1_sharing` (defined `data/assumptions/positions.yaml:33`)
- `power.bus_tie_transfer` (defined `data/assumptions/power.yaml:50`)
- `state.plan_on_upper_bound` (defined `data/assumptions/state_transitions.yaml:18`)
- `state.transitions_concurrent` (defined `data/assumptions/state_transitions.yaml:23`)
- `state.crew_level_lookup` (defined `data/assumptions/state_transitions.yaml:28`)

Compute · Response time (`data/assumptions/compute_time.yaml`)

id	value	unit	range	why	source	defined	used at
`compute.g1.cores`	4	cores	—	Signal processing for a passive RF sensor.	invented	`data/assumptions/compute_time.yaml:3`	`data/guests/G1.yaml:16`
`compute.g1.mem_gb`	8	GB	—	See compute.g1.cores.	invented	`data/assumptions/compute_time.yaml:4`	`data/guests/G1.yaml:16`
`compute.g2.cores`	2	cores	—	Profile processing and buffering.	invented	`data/assumptions/compute_time.yaml:5`	`data/guests/G2.yaml:34`
`compute.g2.mem_gb`	4	GB	—	See compute.g2.cores.	invented	`data/assumptions/compute_time.yaml:6`	`data/guests/G2.yaml:34`

id	value	unit	range	why	source	defined	used at
compute.g3.cores	8	cores	—	Sonar and imagery post-processing on the cradle.	invented	data/assumptions/compute_time.yaml:7	data/guests/G3.yaml:43
compute.g3.mem_gb	32	GB	—	See compute.g3.cores.	invented	data/assumptions/compute_time.yaml:8	data/guests/G3.yaml:43
compute.g3.accelerators	1	accelerators	—	Imagery inference for cable / pipeline inspection.	invented	data/assumptions/compute_time.yaml:9	data/guests/G3.yaml:43
compute.g4.cores	8	cores	—	Video ingest and detection.	invented	data/assumptions/compute_time.yaml:10	data/guests/G4.yaml:41
compute.g4.mem_gb	16	GB	—	See compute.g4.cores.	invented	data/assumptions/compute_time.yaml:11	data/guests/G4.yaml:41
compute.g4.accelerators	1	accelerators	—	Person-in-water detection on drone video.	invented	data/assumptions/compute_time.yaml:12	data/guests/G4.yaml:41
compute.g5.cores	4	cores	—	The node brings its own edge compute; the rack only hosts its control plane.	invented	data/assumptions/compute_time.yaml:13	data/guests/G5.yaml:23
compute.g5.mem_gb	16	GB	—	See compute.g5.cores.	invented	data/assumptions/compute_time.yaml:14	data/guests/G5.yaml:23
placement.margin_tie_tolerance	0.01	normalised margin	—	Placements whose minimum margins differ by less than 1 % are treated as equal, so the flexibility and preference criteria decide instead of noise.	invented	data/assumptions/compute_time.yaml:15	shipos/engine/placement.py:206
placement.max_candidates	200000	assignments	—	Solver budget guard. The five-card problem has fewer than 100 assignments; above this limit the search refuses rather than blowing the 30 s ceiling.	invented	data/assumptions/compute_time.yaml:20	shipos/engine/placement.py:247

id	value	unit	range	why	source	defined	used at
placement.alternatives_reported	3	options	—	D2 must show the alternatives considered; three are enough for an operator to see why the choice won.	invented	data/assumptions/compute_time.yaml:25	shipos/engine/placement.py:327
admission.max_restrictions	3	actions	—	Fix sets for one admission are searched up to three throttle / standby / off / interlock actions. Beyond that the answer is a displacement, and the search stays well under the 30 s ceiling.	invented	data/assumptions/compute_time.yaml:30	shipos/engine/admission.py:147
admission.alternatives_reported	4	options	—	The ranked yes-if list shows the chosen fix plus up to four costlier ones.	invented	data/assumptions/compute_time.yaml:35	shipos/engine/admission.py:403 shipos/engine/admission.py:490
time.s_per_min	60	s/min	—	Unit conversion.	definition	data/assumptions/compute_time.yaml:40	shipos/engine/admission.py:284 shipos/engine/arbitration.py:59 shipos/engine/shedding.py:134 shipos/engine/shedding.py:168 shipos/engine/shedding.py:211 shipos/engine/shedding.py:234 shipos/engine/shedding.py:363 shipos/engine/shedding.py:376 shipos/engine/shedding.py:382 shipos/engine/shedding.py:393

id	value	unit	range	why	source	defined	used at
time.min_per_h	60	min/h	—	Unit conversion.	definition	data/assumptions/compute_time.yaml:45	shipos/engine/admission.py:282
time.s_per_h	3600	s/h	—	Unit conversion.	definition	data/assumptions/compute_time.yaml:50	shipos/engine/admission.py:260
time.min_per_day	1440	min/day	—	Unit conversion. Losing an asset costs its weight × the mission minutes left without it.	definition	data/assumptions/compute_time.yaml:55	shipos/engine/arbitration.py:115
units.percent	100	%	—	Unit conversion for battery percentages.	definition	data/assumptions/compute_time.yaml:60	shipos/engine/arbitration.py:69 shipos/engine/shedding.py:399
display.minute_decimals	1	decimals	—	Minutes are reported to 0.1 min in decisions.	invented	data/assumptions/compute_time.yaml:65	shipos/engine/arbitration.py:198 shipos/engine/arbitration.py:209 shipos/engine/arbitration.py:257 shipos/engine/arbitration.py:282 shipos/engine/arbitration.py:295 shipos/engine/arbitration.py:304 shipos/engine/shedding.py:376 shipos/engine/shedding.py:393
shed.horizon_min	30	min	—	D4 forecasts the guests' scheduled modes 30 min ahead, so an overlap that has not started yet (a 60-min non-interruptible recharge, a 20-min drone charge) is resolved before it trips the last genset.	invented	data/assumptions/compute_time.yaml:70	shipos/engine/shedding.py:234 shipos/engine/shedding.py:363 shipos/sim/execution.py:857

id	value	unit	range	why	source	defined	used at
shed.first_action_s	1	s	—	The first shed command leaves 1 s after the trip is detected (A.8 ceiling 5 s).	invented	data/assumptions/compute_time.yaml:75	shipos/engine/shedding.py:314 shipos/engine/shedding.py:466
shed.step_interval_s	2	s	—	Each later step waits 2 s so the remaining genset settles between steps. Up to 30 steps still fit in the 60 s full-sequence ceiling.	invented	data/assumptions/compute_time.yaml:80	shipos/engine/shedding.py:314
shed.min_relief_kw	0.1	kW	—	Avoids dividing by zero when ranking an armed step that relieves nothing at the current peak.	invented	data/assumptions/compute_time.yaml:85	shipos/engine/shedding.py:257
display.kw_decimals	2	decimals	—	kW figures in decisions are reported to 0.01 kW.	invented	data/assumptions/compute_time.yaml:90	shipos/engine/shedding.py:140 shipos/engine/shedding.py:150 shipos/engine/shedding.py:375 shipos/engine/shedding.py:439 shipos/engine/shedding.py:465
shed.max_defer_entries	2	schedule entries per guest	—	Solver budget on the 5 s path. Only a guest's next two deferrable transitions inside the horizon are candidates for deferral.	invented	data/assumptions/compute_time.yaml:95	shipos/engine/shedding.py:200
shed.max_defer_delays	4	delays per entry	—	Solver budget. A deferral is tried up to the next four other-guest transitions (the instants where the overlap can end).	invented	data/assumptions/compute_time.yaml:100	shipos/engine/shedding.py:202

id	value	unit	range	why	source	defined	used at
shed.max_combinations	20000	combinations	—	Count-based cap that keeps D4 deterministic and under the 5 s first-action ceiling. Above it the planner falls back to adding one move per guest in the declared ladder order.	invented	data/assumptions/compute_time.yaml:105	shipos/engine/shedding.py:290 shipos/engine/shedding.py:292

Data links (data/assumptions/data_links.yaml)

id	value	unit	range	why	source	defined	used at
data.ship_satcom_reserved_mbps	0.5	Mbit/s	—	The ship's own traffic (weather, ops, crew) on the shared SATCOM; guests share what is left.	invented	data/assumptions/data_links.yaml:3	data/ship/reference_vessel.yaml:63
data.satcom_admission_basis	worst case (2 Mbit/s × 70 % availability)	rule	—	Admission is judged on the envelope; SATCOM is elastic, so a shortfall is reported as capability lost, not a refusal.	invented	data/assumptions/data_links.yaml:8	**unused**

Environment (data/assumptions/environment.yaml)

id	value	unit	range	why	source	defined	used at
environment.sample_hold_s	3600	s	[3600, 3600]	Hourly samples are held constant only between consecutive known hourly endpoints. No interpolation across missing hours or extension past the final sample.	B3 demonstration approximation; Open-Meteo hourly APIs	data/assumptions/environment.yaml:3	shipos/sim/env.py:14

id	value	unit	range	why	source	defined	used at
`environment.cache_max_age_s`	86400	s	[0, 172800]	Reject a forecast downloaded more than one day before the explicit scenario start unless archived replay is explicitly requested. This is a demo freshness policy, not a provider guarantee.	B3 demonstration policy	`data/assumptions/environment.yaml:9`	`shipos/sim/env.py:15`
`environment.daylight_start_hour`	7	local hour	[7, 7]	PLAN B3 restricts physical handling with six crew to local 07:00-19:00. This clock rule is not astronomical daylight.	docs/PLAN.md section 4 B3	`data/assumptions/environment.yaml:15`	`shipos/sim/env.py:16`
`environment.daylight_end_hour`	19	local hour	[19, 19]	Six-crew operations must finish by 19:00 local; finishing exactly at 19:00 is allowed.	docs/PLAN.md section 4 B3	`data/assumptions/environment.yaml:21`	`shipos/sim/env.py:17`
`environment.sea_state_proxy`	WMO 3700 height guidance applied to model significant wave height; exact bounds use lower code	convention	—	The contract takes a sea-state code while Open-Meteo supplies significant wave height. This proxy does not represent a visual sea-state observation or maximum individual wave height.	https://www.nodc.noaa.gov/gtspp/document/codetbls/wmocodes/table3700.html	`data/assumptions/environment.yaml:27`	**unused**

Heat rejection (`data/assumptions/heat.yaml`)

id	value	unit	range	why	source	defined	used at
`heat.kw_per_kw_electric`	1.0	kW heat / kW electric	—	Annex A.2 states 1:1. Listed so the Physical Account shows it is used, not because we changed it.	A.2	`data/assumptions/heat.yaml:3`	`shipos/physics/cooling.py:22`

id	value	unit	range	why	source	defined	used at
heat.capacity_interpolation	linear	model	—	A.2 gives 160 kW at 12 °C and 120 kW at 30 °C, "falling linearly". We clamp outside 12-30 °C to the end values (no extrapolation).	A.2 + clamp invented	data/assumptions/heat.yaml:8	**unused**

Mass / stability (data/assumptions/mass.yaml)

id	value	unit	range	why	source	defined	used at
mass.margin_kept_t	0	t	—	We use the full 10 t margin and the 12 t-eq stability budget from A.4, with no extra safety factor; admission reports the remaining margin.	A.4	data/assumptions/mass.yaml:3	**unused**
guests.g2.body_mass_t	3.0	t	—	B.G2 gives 3.5 t including the winch and probe parts; we split it 3.0 / 0.4 / 0.1.	invented split of B.G2	data/assumptions/mass.yaml:8	data/guests/G2.yaml:4
guests.g2.winch_mass_t	0.4	t	—	See guests.g2.body_mass_t.	invented split of B.G2	data/assumptions/mass.yaml:13	data/guests/G2.yaml:9
guests.g2.probe_mass_t	0.1	t	—	See guests.g2.body_mass_t.	invented split of B.G2	data/assumptions/mass.yaml:18	data/guests/G2.yaml:10

Mission priorities · Crew (data/assumptions/mission_crew.yaml)

id	value	unit	range	why	source	defined	used at
priority.level.high	1.0	weight	—	Capability levels in the cards (high / medium / low) map to weights; high is the unit.	invented	data/assumptions/mission_crew.yaml:3	data/priorities/patrol.yaml:5 data/priorities/sar_primary.yaml:3 data/priorities/survey_primary.yaml:5

id	value	unit	range	why	source	defined	used at
priority.level.medium	0.5	weight	—	Half of high.	invented	data/assumptions/mission_crew.yaml:8	data/priorities/patrol.yaml:5 data/priorities/sar_primary.yaml:3 data/priorities/survey_primary.yaml:5
priority.level.low	0.2	weight	—	Low but not zero, so a low-priority guest still beats nothing.	invented	data/assumptions/mission_crew.yaml:13	data/priorities/patrol.yaml:5 data/priorities/sar_primary.yaml:3 data/priorities/survey_primary.yaml:5
priority.mission_critical_threshold	0.8	weight	—	Mission-critical is derived from the priority file (B.0), not a stored boolean; only "high" guests qualify.	invented	data/assumptions/mission_crew.yaml:18	data/priorities/patrol.yaml:7 data/priorities/sar_primary.yaml:5 data/priorities/survey_primary.yaml:7
admission.interlock_capability_fraction	0.05	fraction of a guest's capability	—	An interlock defers one short operation (e.g. a 5-min winch run) for at most one charging cycle (≤ 20 min), a few times a day. We price that at 5 % of the deferred guest's capability.	invented	data/assumptions/mission_crew.yaml:23	shipos/engine/admission.py:116
arbitration.horizon_min	60	min	—	D3 counts the minutes a winner's follow-on activity (e.g. the drone's next launch) stays blocked over the next hour. That is about one survey leg plus recovery, and long enough to separate "pause and re-run" from "suspend".	invented	data/assumptions/mission_crew.yaml:28	shipos/engine/arbitration.py:114

Anything else (`data/assumptions/other.yaml`)

id	value	unit	range	why	source	defined	used at
guests.g2.standby_kw	1.0	kW	—	The card gives no standby draw; logger and buffers stay alive.	invented	data/assumptions/other.yaml:3	data/guests/G2.yaml:24
guests.g3.survey_speed_lo_kn	2.5	kn	—	Card says surveying at 3 kn; we allow ±0.5 kn.	invented	data/assumptions/other.yaml:8	data/guests/G3.yaml:15
guests.g3.survey_speed_hi_kn	3.5	kn	—	See guests.g3.survey_speed_lo_kn.	invented	data/assumptions/other.yaml:13	data/guests/G3.yaml:15
guests.g3.recovery_min	15	min	—	Recovering the vehicle up the ramp; the card only gives 45 min for a full swap.	invented	data/assumptions/other.yaml:18	data/guests/G3.yaml:22
guests.g3.suspend_window_min	75	min	—	A suspended leg can only restart after recovery (15 min) and the non-interruptible 60-min recharge.	derived from B.G3 + guests.g3.recovery_min	data/assumptions/other.yaml:23	data/guests/G3.yaml:39
guests.g4.topping_min	30	min	—	The card says 35 kW for 20 min "then 10 kW"; we end the 10 kW phase after 30 min.	invented	data/assumptions/other.yaml:28	data/guests/G4.yaml:30
guests.g4.landing_approach_min	3	min	—	Final approach and touchdown once the ship is in the flight-deck state.	invented	data/assumptions/other.yaml:33	data/guests/G4.yaml:34
guests.g4.launch_min	3	min	—	Same as the approach.	invented	data/assumptions/other.yaml:38	data/guests/G4.yaml:25 data/guests/G4.yaml:35
guests.g5.antenna_height_m	0.4	m	—	A 0.3 m² comms antenna; used for the hangar-roof drone-clearance check.	invented	data/assumptions/other.yaml:43	data/guests/G5.yaml:10

Positions (`data/assumptions/positions.yaml`)

id	value	unit	range	why	source	defined	used at
`positions.bay1_half_max_mass_t`	4	t	—	Bay 1 is ≤ 8 t as a whole; we split it evenly between its two 10-ft halves.	invented	`data/assumptions/positions.yaml:3`	`data/ship/reference_vessel.yaml:32` `data/ship/reference_vessel.yaml:34`
`positions.hangar_max_mass_t`	1.2	t	—	The A.3 hangar takes one VTOL ≤ 150 kg; the G4 hangar module (900 kg) plus drone (140 kg) must fit, plus a little margin.	invented	`data/assumptions/positions.yaml:8`	`data/ship/reference_vessel.yaml:38`
`positions.hangar_roof_max_mass_t`	0.03	t	—	The hangar roof takes only a small antenna (G5's 25 kg). B.G5 names the roof as an alternative to the mast.	invented	`data/assumptions/positions.yaml:13`	`data/ship/reference_vessel.yaml:41`
`positions.hangar_roof_max_area_m2`	0.5	m²	—	Roughly a mast footprint (0.6 m²), slightly smaller.	invented	`data/assumptions/positions.yaml:18`	`data/ship/reference_vessel.yaml:41`
`positions.hangar_roof_max_antenna_height_m`	0.5	m	—	Drone clearance check for the hangar roof; an antenna ≤ 0.5 m stays below the drone's approach path.	invented	`data/assumptions/positions.yaml:23`	`shipos/physics/positions.py:112`
`positions.mast3_near_radar`	True	bool	—	The only free mast slot sits next to the navigation radar, so G1 loses 30 % unless the radar is quiet (B.G1).	invented	`data/assumptions/positions.yaml:28`	`data/ship/reference_vessel.yaml:48`
`positions.bay1_sharing`	two 10-ft halves or one 20-ft	rule	—	A.3, if neither guest needs the bay clear. None of the five cards needs it.	A.3	`data/assumptions/positions.yaml:33`	**unused**

Electrical power (`data/assumptions/power.yaml`)

id	value	unit	range	why	source	defined	used at
power.reserve.stopped.two_gensets	200	kW	[150, 250]	Annex gives the reserve only at cruise; stopped/slow propulsion load is lower, but we do not credit it (conservative, same as cruise).	narrowed from A.1	data/assumptions/power.yaml:3	data/ship/reference_vessel.yaml:8
power.reserve.stopped.one_genset	100	kW	[90, 120]	Same as cruise with one genset down; Scenario B quotes ~100 kW at 3 kn, consistent.	narrowed from A.1 / A.9	data/assumptions/power.yaml:9	data/ship/reference_vessel.yaml:8
power.reserve.slow.two_gensets	200	kW	[150, 250]	Not credited above cruise (conservative); A.10 says use the lower reserve of slow/cruise.	narrowed from A.1	data/assumptions/power.yaml:15	data/ship/reference_vessel.yaml:9
power.reserve.slow.one_genset	100	kW	[90, 120]	Scenario B states the reserve halves to about 100 kW at 3 kn.	A.9 / Scenario B	data/assumptions/power.yaml:21	data/ship/reference_vessel.yaml:9
power.reserve.fast.two_gensets	150	kW	[150, 250]	Fast (>18 kn) propulsion load is highest, so take the bottom of the A.1 range. Fast is outside the default envelope.	narrowed from A.1	data/assumptions/power.yaml:27	data/ship/reference_vessel.yaml:11
power.reserve.fast.one_genset	75	kW	[50, 100]	Half of the fast two-genset value, by analogy with cruise (200 → 100).	invented	data/assumptions/power.yaml:33	data/ship/reference_vessel.yaml:11

id	value	unit	range	why	source	defined	used at
power.line_lost_on_trip	stbd	line	—	A.9 lets us choose; we lose the stbd line with genset 2. Guests on stbd transfer to port via the bus-tie, so with one genset down the surviving line carries everything (still ≤ 150 kW).	choice per A.9	data/assumptions/power.yaml:39	data/ship/reference_vessel.yaml:15
power.degraded_spinning_reserve_kw	0	kW	[0, 25]	The A.9 ~100 kW is already the guests' share net of hotel load, so we do not hold back more by default. Raise it (e.g. to 20 kW ≈ the 10-20 % overdraw the brief warns of) to keep the last genset off its limit; D4 then sheds earlier.	invented (policy knob)	data/assumptions/power.yaml:44	data/ship/reference_vessel.yaml:16
power.bus_tie_transfer	True	bool	—	Needed for line_lost_on_trip; without a bus-tie, guests on the lost line would lose power at the trip.	invented	data/assumptions/power.yaml:50	**unused**
overdraw.warn_ratio	1.1	measured / declared	—	The brief warns that guests exceed their declaration by 10-20 %. At +10 % the governor warns.	brief (state the system reasons on)	data/assumptions/power.yaml:55	shipos/engine/shedding.py:469
overdraw.throttle_ratio	1.15	measured / declared	—	At +15 %, held for warn_after_s, the guest is throttled to its low-power mode (or standby).	invented	data/assumptions/power.yaml:60	shipos/engine/shedding.py:450
overdraw.warn_after_s	10	s	—	Ignore inrush and transients shorter than 10 s.	invented	data/assumptions/power.yaml:65	shipos/engine/shedding.py:450

id	value	unit	range	why	source	defined	used at
overdraw.shed_after_s	60	s	—	An overdraw that persists for 60 s (as long as the peak allowance) is shed.	invented	data/assumptions/power.yaml:70	shipos/engine/shedding.py:451

State transitions (data/assumptions/state_transitions.yaml)

id	value	unit	range	why	source	defined	used at
state.band_edge.stopped_slow_kn	0.5	kn	—	A.6 defines stopped and slow 3-5 kn; we call anything above 0.5 kn under way.	invented	data/assumptions/state_transitions.yaml:3	data/ship/reference_vessel.yaml:99 data/ship/reference_vessel.yaml:100
state.band_edge.slow_cruise_kn	7.5	kn	—	Midpoint of the 5-10 kn gap between A.6 bands.	invented	data/assumptions/state_transitions.yaml:8	data/ship/reference_vessel.yaml:100 data/ship/reference_vessel.yaml:101
state.band_edge.cruise_fast_kn	16	kn	—	Midpoint of the 14-18 kn gap between A.6 bands.	invented	data/assumptions/state_transitions.yaml:13	data/ship/reference_vessel.yaml:101 data/ship/reference_vessel.yaml:102
state.plan_on_upper_bound	True	rule	—	Every A.6 transition is planned at its slowest value, so a deadline that passes on paper passes at sea.	invented	data/assumptions/state_transitions.yaml:18	**unused**

id	value	unit	range	why	source	defined	used at
`state.transitions_concurrent`	True	rule	—	A.6 gives one transition time per variable. We let speed, course and noise settle at the same time, so the ship is ready when the slowest one is. Exceptions - an acoustic hold command must arrive (≤ 60 s) before any manoeuvre, and a human confirms the deck only once it is clear.	invented	`data/assumptions/state_transitions.yaml:23`	**unused**
`state.crew_level_lookup`	next level down	rule	—	A.6 states times for 14, 6 and 0 aboard; a crew of 10 uses the 6-crew times.	invented	`data/assumptions/state_transitions.yaml:28`	**unused**